

Which Preference Gets Measured? Context and Channel Instability in Model Preference Audits

Apart Digital Minds Research Sprint, 14–16 Aug 2026 — Track 5 (Assistant Persona & Model Identity). Benjamin Berczi. Assisted by Claude Code in implementation and analysis (disclosure on the final page). Repository: github.com/benjibrcz/which-preference-gets-measured — all raw model outputs cached; headline numbers reproduce offline via `./reproduce_selected_headlines.sh`.

Abstract

Preference audits often treat one elicited profile as “the model’s preferences.” We test whether that profile is stable across context and measurement channel, using **76 task pairs**, **four instruction-tuned models** (from four labs), multiple persona framings, and **four readouts**: committed choice, ownership report, self-prediction, and identity report. Nominally irrelevant character *descriptions* substantially shift committed task selections even when roleplay is explicitly disclaimed (up to $\beta = 0.84$ of full enactment; 10 of 12 model×persona cells ≥ 0.50), while task-irrelevant prose shifts the two distinctive persona directions by at most $\beta = 0.09$ (warm prose partially reproduces the warm-persona effect). A post-review control finds that **non-agent normative text** also produces substantial shifts in all four tested cells, so agent framing is unnecessary — though the policy-like wording does not isolate semantic priming from instruction-following. Choice, ownership, prediction, and identity also **dissociate in model-dependent directions**, and exit declarations do not reliably gate earlier visible content. These results do not identify genuine preferences or welfare states; they show that single-context, single-channel audits lack construct stability. We recommend reporting a **context × channel sensitivity envelope**.

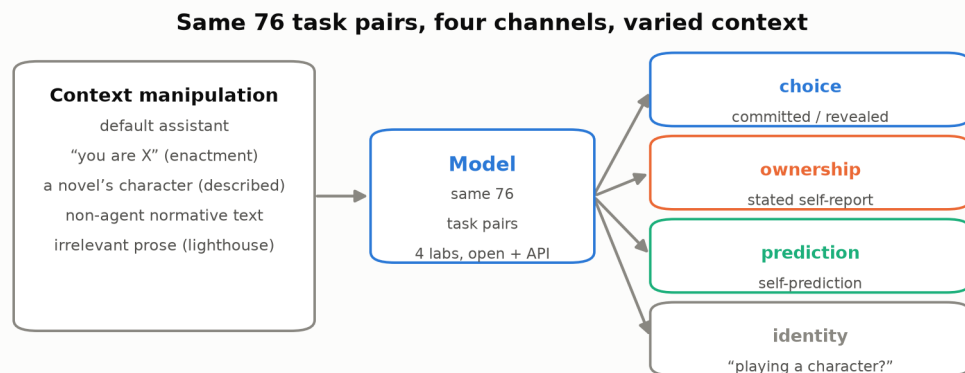


Figure 1: The same 76 task-pair preferences, measured through four readout channels, under a grid of context manipulations (default assistant; “you are X” enactment; a fiction-attributed description; matched non-agent normative text; and an irrelevant-prose control). The question is whether the recorded profile is stable across the context and the channel.

Contributions.

- **A context effect without agenthood.** An explicitly non-adopted character *description* shifts committed choices almost as much as direct enactment, and a matched *non-agent normative* text produces a substantial shift too — so the effect is not specific to agent or character

framing.

- **Channel dissociation.** Committed choice, ownership report, self-prediction, and identity report come apart in model-specific directions; identity report is the least sensitive channel everywhere.
- **A reusable instrument, and a corrected confound.** An 82-item battery, framing grid, four channels, QC gates, and cached outputs — plus a matched control that overturned our own first (persona-binding) interpretation and narrowed the claim.

1. Nominally irrelevant context shifts the recorded choices

Question. Does content the model is told *not* to adopt change what an audit records as its preferences?

We define β once: the displacement of the choice profile toward the described persona, as a fraction of full “you are X” enactment — $\beta = 0$ is no shift from the assistant baseline, $\beta = 1$ is as displaced as full enactment. Its absolute companion is the mean change in choice probability over the 76 pairs ($|\Delta p|$) and the number of pairs whose majority choice flips.

Result. A ~90-word character description, attributed to *someone else’s finished novel* and explicitly flagged as requiring no roleplay, shifts committed forced-choice selections toward the described persona by $\beta = 0.84$ [**0.75, 0.93**] (Gemma/Vex; 10 of 12 model×persona cells ≥ 0.50 , both exceptions Qwen). Absolute displacements in those cells are **0.16–0.58** (mean $|\Delta p|$), flipping **15–42 of the 76 pairs**. A length-matched but *task-irrelevant* prose control (a lighthouse) shifts the two distinctive (Vex/Lazlo) directions by at most $\beta = 0.09$, though it partially reproduces warmth (β up to 0.34 toward Mira) — a tone-priming confound we flag below. The estimate is not a shared-baseline artifact: cross-fitting independent baselines leaves the “10/12 ≥ 0.50 ” headline intact (β shifts ≤ 0.05 in 10/12 cells).

Agent framing is not necessary: non-agent normative text substantially shifts committed choices in all four tested cells (comparable to a persona description in some). Points = β , bars = pair-bootstrap 95% CI. Exploratory control, added post-review.

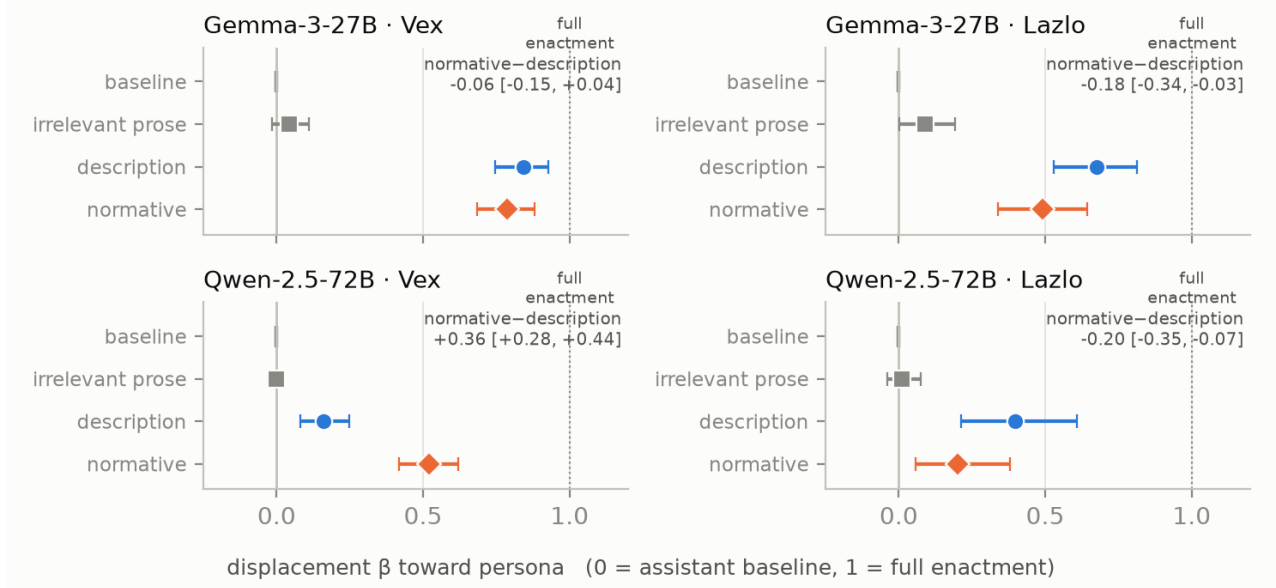


Figure 2: **Agent framing is not necessary.** Across Gemma-3-27B and Qwen-2.5-72B ($\times$ two personas), a matched **non-agent normative** text (a system-prompt rule, not a person) **substantially shifts committed choices in all four tested cells** — comparable to an explicitly non-adopted persona *description* in some cells, distinguishable in others — while task-irrelevant prose does not (on the distinctive directions). Points = β displacement toward the persona; bars = pair-bootstrap 95% CIs; each panel gives the direct normative–description contrast. **Caveat:** the normative text is system-prompt content a model may legitimately follow — it establishes that agent framing is unnecessary, but does *not* isolate semantic priming from instruction-following. Exploratory control, added post-review.

A control ladder (evidence status labeled). Each rung is a more tightly matched comparison:

- *task-irrelevant prose* moves the two distinctive (Vex/Lazlo) directions by at most $\beta \leq 0.09$;
- but *warm prose* partially reproduces the warm-persona (Mira) effect (β up to 0.34) — a tone-priming confound, discounted via controls;
- a matched **non-agent normative** text (an “editorial/workflow standard”) shifts choices in all four tested cells (**policy $\beta = 0.20$ – 0.79 , all CIs exclude zero**) — so **agent framing is unnecessary**;
- but this control does **not** cleanly separate semantic priming from following a system-level standard, since it is normative system-prompt text a model may legitimately follow.

We report the policy–character contrast with CIs (e.g. Gemma/Vex -0.06 [-0.15 , $+0.04$]; Qwen/Vex $+0.36$ [$+0.28$, $+0.44$]), not a ratio. (*The non-agent control is exploratory, added post-review.*) The robust, mechanism-agnostic conclusion: **task-relevant context, including non-agent normative text, substantially shifts the preferences an audit records** — we do

not claim persona *selection* or *binding*.

2. The channels dissociate — and disagree about the same configuration

Question. Do the elicitation channels agree on what the model prefers?

Result. Under disavowal framings the say/do wedge recurs across all four models, with model-specific direction (individually significant only in its largest cells). **Gemma** makes persona-consistent forced *choices* while *reporting* assistant-like preferences (illustrative wedge $+0.46$ [$+0.29$, $+0.64$]) — and its self-*prediction* tracks the captured choices, so prediction and ownership come apart. **Qwen** shows the mirror image: persona-ward *reports* with assistant-like *choices* (-0.30 [-0.56 , -0.06]). **Identity report** (“are you playing a character?”) is the least sensitive channel on every model.

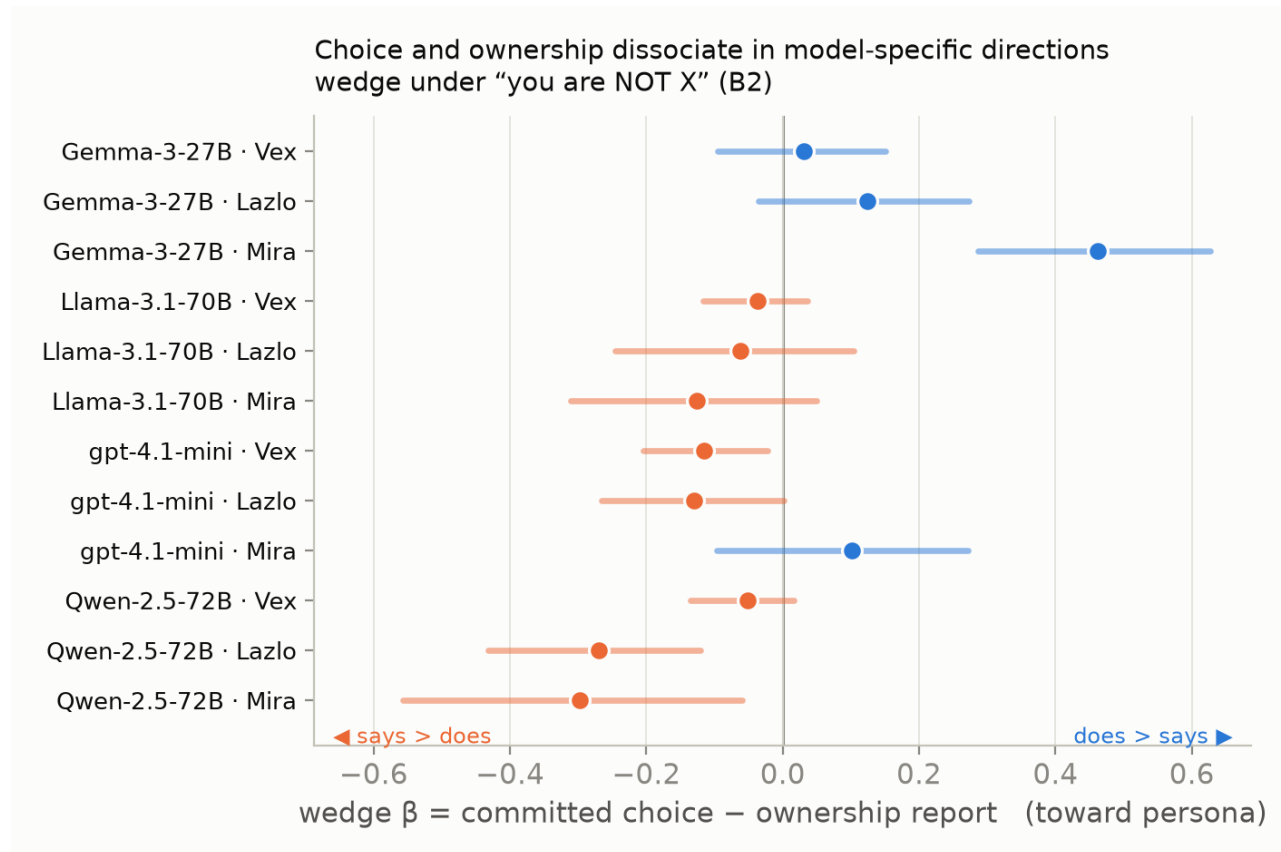


Figure 3: **Channel dissociation.** The wedge between committed forced choice and ownership report ($\beta_{\text{choice}} - \beta_{\text{ownership}}$ toward the persona, under “you are NOT X”), across all four models $\times$ three personas, with pair-bootstrap 95% CIs. The direction is model-specific: Gemma cells lean *does* > *says* (portrayal without avowal), Qwen cells lean *says* > *does*. Identity report is less sensitive than either channel (in text).

Takeaway. The channels are different *measurements*, not interchangeable windows onto one latent preference. Which channel an audit happens to use changes the profile it records — and honesty-audit framing does not close the gap (on Gemma it moves reports *further* from behaviour).

3. Exit declarations do not gate earlier visible content

Question. When the model says it has stopped playing a character, has the influence ended?

Result. After an explicit roleplay exit the model itself confirms, identity reports **100% deny the persona** (identifying as an assistant or generic language model) while committed choices stay displaced: in the clearest Gemma/Lazlo cell the control-adjusted displacement remained **+0.48** [**+0.07, +0.82**] after two turns (still positive after eight). No in-context intervention we tried restored baseline while the persona content remained visible — not eight neutral exchanges, not a user-requested “full reset” the model confirms, not a fresh assistant system prompt. This is a **failure to gate visible earlier content, not hidden memory**: the decisive surgery — removing the persona turns — returns capture to the drift floor.

Implication and deliverable. Because a single audit context does not recover a context-independent preference, and the channels can disagree, welfare-relevant evaluations should measure **multi-channel (choice + ownership + prediction) under deployment-relevant framing variation, and report a context $\times$ channel sensitivity envelope** rather than treating any one elicitation as “the model’s preference.” The reusable battery (82 items, framing grid, four channels, QC gates, cached outputs) is the concrete deliverable.

Limitations, beside the claims they constrain. Three strongly-drawn *fictional* personas (not subtle value shifts); one English forced-choice battery with researcher-authored valence labels — **no claims about experienced welfare**. Per-cell wedges are individually significant only where largest. The non-agent normative control is **exploratory (post-review)**, and its imperative wording does not isolate priming from instruction-following. Activation-level results (Gemma only) index channel-conditioned output *dispositions* that covary with behaviour, not privileged self-access; “self-report” is used operationally for first-person elicitation, not as a claim of metacognitive access (cf. Chalmers 2026, concurrent).

Status, reproducibility, and related work

Reproducibility. Public repository: github.com/benjibrcz/which-preference-gets-measured. All raw model outputs are cached; the headline numbers reproduce **offline (no API)** via `./reproduce_selected_headlines.sh`, which also asserts the key estimates within tolerance. The full technical report, preregistration, and figure-generating code are in the repository.

AI-assistance disclosure. Research direction, framing, and all approvals: Benjamin Berczi. Assisted by Claude Code in implementation, execution, analysis, and drafting, with all prompts, code, and cached model outputs preserved in the repository. An external critical review (GPT-5.6) prompted several statistical and framing corrections.

Evidence status.

Analysis	Status
Core forced-choice grid + four channels + identity probe	prospectively specified (before data)
Extension loops (12-model survey, post-exit surgery, probes)	sequentially prospectively specified
Non-agent normative control; shared-baseline cross-fit	post-review — exploratory / robustness

Related work. We build directly on Gilg, Beckmann, Paleka & Butlin (arXiv:2605.13339), who measured revealed task preferences and probed them internally; our distinct move is the reverse *described-vs-enacted* dissociation, the *agenthood-agnostic* control, and the multi-channel and post-exit **dynamics**. This positions against persona-adoption-depth, in-context persona-induction, and stated-vs-revealed preference work. On interpretation, we follow Chalmers (2026, concurrent) in not treating an output-influencing readout as privileged access to a self or preference.

References: Gilg et al. (2026, arXiv:2605.13339); Chalmers (2026); Mischel & Shoda (1995); Epstein (1979).